

Scalability & Future Plan

Date	05 July 2026
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the OptiCrop solution can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also captures the team's roadmap for enhancing and evolving the platform beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Render free-tier cold start: the server spins down after inactivity, causing an 8–12 second delay for the first request	Farmers may experience a slow initial load, reducing trust in the tool during demonstrations or first-time use	High
2	Single static ML model: the crop recommendation models are trained once on a fixed dataset and do not learn from new real-world farm inputs	Predictions may become less accurate over time as climate patterns shift or new crop varieties are introduced	High
3	No user authentication or data persistence: users cannot save previous analyses or track crop recommendations across planting seasons	Farmers must re-enter inputs every session; no historical trend analysis is possible	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Render free tier – ~50 concurrent users before performance degradation	Upgrade to Render paid tier or migrate to AWS EC2 with auto-scaling; use gunicorn with multiple workers to handle 1,000+ concurrent users
2	Data Storage	CSV-based static dataset (2,200 records, 22 crops); no database	Migrate to PostgreSQL on AWS RDS to store user-submitted soil profiles, historical recommendations, and regional farm data for model retraining

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
3	Performance	Single-process Flask; ML models loaded in memory at startup; avg response 1.4 sec	Implement Redis caching for repeated similar inputs; explore model quantization and TensorFlow Lite for faster mobile inference
4	Security	No authentication; HTTPS via Render; no user data stored	Add OAuth2-based user login (Google/email); implement rate limiting on the /predict endpoint; add input sanitization and CSRF protection

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Authentication & Farm History — Add Google/email login; allow farmers to save and review past crop analyses across planting seasons	3–4 months	Higher user retention; farmers can track seasonal trends and compare year-on-year crop performance
Phase 3	Real-Time Weather & Soil API Integration — Connect to weather APIs (OpenWeatherMap) and ICAR soil databases to auto-populate temperature, humidity, and rainfall fields based on farm GPS location	5–7 months	Reduces manual input errors; makes the tool usable by farmers who lack soil lab reports
Phase 4	Multi-Language Support — Add Telugu and Hindi language interfaces to reach rural farming communities in Andhra Pradesh, Telangana, and other Hindi-speaking states	7–9 months	Dramatically widens the user base to non-English-speaking farmers; addresses the digital literacy gap
Phase 5	Yield Prediction & Market Price Forecasting — Extend the ML pipeline to predict expected yield quantity and integrate commodity market price data to recommend the most profitable crop, not just the most suitable one	10–12 months	Transforms OptiCrop from a crop selector into a full farm decision-intelligence platform; directly increases farmer income